

Year 2 Assessed Problems

Semester 2

Assessed Problems

SOLUTIONS TO BE SUBMITTED

ON CANVAS BY

Wednesday 25 March

at 17:00

Y2 Electromagnetism 2

Assessed problem sheet 3 (weeks 6–9)

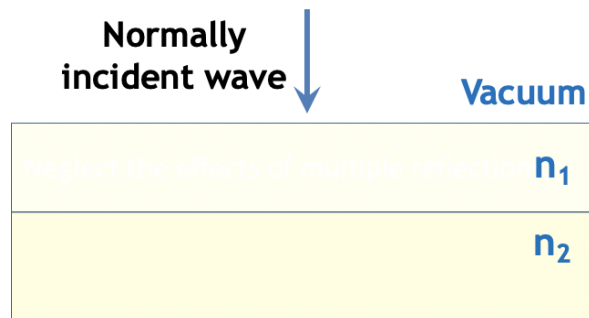
1) Consider a circularly polarised plane wave propagating in free space in the negative z direction. The electric field is

$$\vec{E}(z, t) = \vec{e}_x E_0 \cos(\omega t + kz) - \vec{e}_y E_0 \sin(\omega t + kz),$$

where $\vec{e}_x$ and $\vec{e}_y$ are unit vectors in the x and y directions, respectively.

- Establish the relation between ω and k . [1]
- Determine the magnetic field $\vec{B}(z, t)$ in this wave in terms of E_0 , ω and k . [2]
- Find the magnitude and direction of the energy flux. Does the energy flux depend on coordinates and time? [2]

2) Consider a plane electromagnetic wave propagating in vacuum, normally incident on a structure consisting of two layers of dielectric materials, with refractive indices n_1 and n_2 . At a fixed value of n_2 , which value of n_1 leads to maximum transmittance from vacuum into the material n_2 ? Find the maximum transmittance for $n_2 = 1.44$, and compare it to the case when the intermediate layer n_1 is absent. Neglect the effects of multiple reflections. [5]



Eigenphysics 2025-2026: Assessed Problem 3

State whether each of the following sets is a vector space or not. If it is not a vector space, state one axiom that is not obeyed and explain why it is not obeyed.

- (a) Real symmetric $n \times n$ matrices i.e. matrices which satisfy $A_{ij} = A_{ji}$.
- (b) Real orthogonal $n \times n$ matrices i.e. matrices which satisfy $AA^T = A^T A = I$.
- (c) Real $n \times n$ matrices which commute with a given matrix X i.e. matrices which satisfy $AX = XA$.
- (d) Real $n \times n$ matrices, A , which have zero determinant.
- (e) Solutions of $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 1$.
- (f) Solutions of $\frac{dy}{dx} \frac{d^2 y}{dx^2} = 0$.
- (g) Polynomials with $p'(0) = 1$.
- (h) Polynomials with $\int_0^1 p(x)dx = 0$
- (i) Convergent real sequences $\{u_n\}$.
- (j) Divergent real sequences $\{u_n\}$.

Assessed Problems 3

Richard Mason
Foundations of Data Science

Problem 3. Consider two datasets which both follow the exponential distribution

$$\begin{aligned}P(x|\lambda) &= \lambda_x e^{-\lambda_x x} & x \geq 0 \\P(y|\lambda) &= \lambda_y e^{-\lambda_y y} & y \geq 0\end{aligned}$$

Let $\bar{x}$ denote the sample mean for one data set with N points, and $\bar{y}$ represent the second dataset with M points.

1. Employ the central limit theorem to calculate the asymptotic distribution of the sample means. [3]
2. Construct a Hypothesis test based upon these that tests whether $\lambda_x = \lambda_y$ which follows a standard Normal distribution asymptotically. [4]

Hint: use the grand mean to estimate $1/\lambda$ defined by

$$\bar{G} = \frac{N}{N+M}\bar{x} + \frac{M}{N+M}\bar{y}$$

and you may use equivariance to estimate $1/\lambda^2$.

3. If $\bar{x} = 2.4$ and $N = 20$ and $\bar{y} = 2.8$ with $M = 18$, does this provide evidence that the two parameters are the same? The two-sided acceptance region for a standard Normal at $\alpha = 0.05$ is $[-1.96, 1.96]$. [3]